

Table A19: **List of substitution datasets** See the reference file in the GitHub repo for other info (UniProt ID, taxon, DOI, and more assay details).

Dataset	Reference
β -Lactamase	Jacquier et al. [2013]
β -Lactamase	Stiffler et al. [2015]
β -Lactamase	Firnberg et al. [2014]
β -Lactamase	Deng et al. [2012]
β -Lactamase VIM-2	Chen et al. [2020]
β -Glucosidase	Romero et al. [2015]
AAV	Sinai et al. [2021]
ACE2	Chan et al. [2020]
ADRB2	Jones et al. [2020]
APH(3')II, neo	Melnikov et al. [2014]
APP	Seuma et al. [2021]
Activation-induced deaminase	Gajula et al. [2014]
Aliphatic amidase	Wrenbeck et al. [2017]
Alpha-synuclein	Newberry et al. [2020]
Amyloid β	Gray et al. [2019]
Amyloid β	Seuma et al. [2022]
Ancestral spleen tyrosine kinase	Hobbs et al. [2022]
Anti-CRISPR protein AcrIIA4	Stadelmann et al. [2021]
Antitoxin ParD3	Ding et al. [2023]
Antitoxin ParD3	Aakre et al. [2015]
Arrestin-1	Ostermaier et al. [2014]
BRCA1	Findlay et al. [2018]
BRCA2	Erwood et al. [2022]
CALM1	Weile et al. [2017]
CARD11	Meitlis et al. [2020]
CASP3	Roychowdhury and Romero [2022]
CASP7	Roychowdhury and Romero [2022]
CBS (cystathionine beta-synthase)	Sun et al. [2020]
CCR5	Gill et al. [2023]
CD19	Klesmith et al. [2019]
CVB3 capsid	Mattenberger et al. [2021]
CXCR4	Gill et al. [2023]
Chalcone synthase	Wrenbeck et al. [2019]
Cytochrome P450 2C9	Amorosi et al. [2021]
Cytochrome P450 2C9	Amorosi et al. [2021]
D-amino acid oxidase	Vanella et al. [2023]
DHFR reductase	Nguyen et al. [2023a]
DHFR reductase	Thompson et al. [2020]
DNA methylase HaeIII	Rockah-Shmuel et al. [2015]
Dengue virus NS5	Suphatrakul et al. [2023]
Dlg4, (PSD95_PDZ3)	McLaughlin et al. [2012]
EfrC	Meier et al. [2023]
EfrD	Meier et al. [2023]
EnvZ	Ghose et al. [2023]
ErbB2 membrane domain	Elazar et al. [2016]
EstA	Nutschel et al. [2020]
GAL4	Kitzman et al. [2015]
GB1	Wu et al. [2016]
GB1	Olson et al. [2014]
GDI1	Silverstein et al. [2021]
GFP	Sarkisyan et al. [2016]
GMR (aacC1)	Dandage et al. [2018]
GRB2-SH3	Faure et al. [2022]
Gcn4	Staller et al. [2018]
Glucokinase regulatory protein	Gersing et al. [2023]
Glucokinase regulatory protein	Gersing et al. [2022]
Glycophorin A membrane domain	Elazar et al. [2016]
Green fluorescent protein amacGFP	Gonzalez Somermeyer et al. [2022]
Green fluorescent protein cgreGFP	Gonzalez Somermeyer et al. [2022]
Green fluorescent protein ppluGFP2	Gonzalez Somermeyer et al. [2022]

Table A19: (continued)

Dataset	Reference
HIV env	Duenas-Decamp et al. [2016]
HIV env	Haddox et al. [2018]
HIV env (BF520)	Haddox et al. [2018]
HIV env (BG505)	Haddox et al. [2018]
HIV rev	Fernandes et al. [2016]
HIV tat	Fernandes et al. [2016]
HMG-CoA reductase	Jiang [2019]
HRAS	Bandaru et al. [2017]
HSP82	Flynn et al. [2020]
HSP82	Mishra et al. [2016]
Hsp90	Hietpas et al. [2011]
Hydroxymethylbilane synthase	van Loggerenberg et al. [2023]
IGP dehydratase (HIS3)	Pokusaeva et al. [2019]
InfA	Kelsic et al. [2016]
Influenza H3N2 neuraminidase	Lei et al. [2023]
Influenza M1 matrix protein	Hom et al. [2019]
Influenza RNA polymerase PB1	Li et al. [2023]
Influenza hemagglutinin	Thyagarajan and Bloom [2014]
Influenza hemagglutinin	Doud and Bloom [2016]
Influenza hemagglutinin	Wu et al. [2014]
Influenza hemagglutinin	Lee et al. [2018]
Influenza neuraminidase	Jiang et al. [2016]
Influenza nucleoprotein	Bloom [2014]
Influenza nucleoprotein	Doud et al. [2015]
Influenza nucleoprotein	Doud and Bloom [2016]
Influenza polymerase acidic protein	Wu et al. [2015]
Influenza polymerase basic protein 2	Soh et al. [2019]
KCNE1	Muhammad et al. [2023]
KCNH2	Kozek et al. [2020]
KCNJ2	Coyote-Maestas et al. [2022]
KRAS	Weng et al. [2022]
KRAS	Ursu et al. [2022]
L-selectin	Elazar et al. [2016]
LGK (levoglucosan kinase)	Wrenbeck et al. [2019]
LGK (levoglucosan kinase)	Klesmith et al. [2015]
LamB	Andrews and Fields [2020]
Leucine-rich repeat protein SHOC-2	Kwon et al. [2022]
MAPK1	Brenan et al. [2016]
MET kinase	Estevam et al. [2023]
MPL	Bridgford et al. [2020]
MSH2	Jia et al. [2021]
MTHFR reductase	Weile et al. [2021]
MlaC	MacRae et al. [2023]
NPC intracellular cholesterol transporter	Erwood et al. [2022]
NPC intracellular cholesterol transporter	Erwood et al. [2022]
NS5A	Qi et al. [2014]
NUDT15	Suiter et al. [2020]
OCT1 (SLC22A1)	Yee et al. [2023]
Ornithine transcarbamylase	Lo et al. [2023]
p53	Giacomelli et al. [2018]
p53	Kotler et al. [2018]
PAB1	Melamed et al. [2013]
PPARG	UK Monogenic Diabetes Consortium et al. [2016]
PSD95-PDZ3	Faure et al. [2022]
PTEN	Matreyek et al. [2021]
PTEN	Mighell et al. [2018]
Parkin	Clausen et al. [2023]
Phosphoserine aminotransferase	Xie et al. [2023]
Phototropin	Chen et al. [2023]
Pilin (PilE)	Kennouche et al. [2019]
Plasminogen activator inhibitor-1	Huttinger et al. [2021]

Table A19: (continued)

Dataset	Reference
Protein phosphatase 1D	Miller et al. [2022]
RAF oncogene	Zinkus-Boltz et al. [2019]
RNAse III (rnc)	Weeks and Ostermeier [2023]
Rhodopsin	Wan et al. [2019]
SARS-CoV-2 Mpro	Flynn et al. [2022]
SARS-CoV-2 Spike RBD	Tan et al. [2023]
SARS-CoV2 Spike RBD	Starr et al. [2020]
SCN5A	Glazer et al. [2020]
SOX17	Veerapandian et al. [2018]
SOX2	Veerapandian et al. [2018]
SRC	Ahler et al. [2019]
SUMO-conjugating enzyme UBC9	Weile et al. [2017]
Small ubiquitin-related modifier 1	Weile et al. [2017]
Sodium-dependent serotonin transporter	Young et al. [2021]
Src	Chakraborty et al. [2021]
Src	Nguyen et al. [2023b]
Streptococcus pyogenes Cas9	Spencer and Zhang [2017]
TARDBP	Bolognesi et al. [2019]
TIM Barrel (S. solfataricus)	Chan et al. [2017]
TIM Barrel (T. maritima)	Chan et al. [2017]
TIM Barrel (T. thermophilus)	Chan et al. [2017]
Thiamin pyrophosphokinase 1	Weile et al. [2017]
Thiopurine S-methyltransferase (TPMT)	Matreyek et al. [2018]
Toxin CcdB	Tripathi et al. [2016]
Toxin CcdB	Adkar et al. [2012]
Tsuboyama multi-DMS	Tsuboyama et al. [2023]
Ube4b	Starita et al. [2013]
Ubiquitin	Roscoe et al. [2013]
Ubiquitin	Roscoe and Bolon [2014]
Ubiquitin	Mavor et al. [2016]
VKORC1	Chiasson et al. [2020]
VKORC1	Chiasson et al. [2020]
YAP1	Araya et al. [2012]
Zika virus env	Sourisseau et al. [2019]

Dataset	Reference
β -Lactamase	Gonzalez et al. [2019]
AAV	Sinai et al. [2021]
Chorismate mutase (CM)	Russ et al. [2020]
IGP dehydratase (HIS3)	Pokusaeva et al. [2019]
Kir2.1	Macdonald et al. [2023]
MtrA	Campbell et al. [2022]
p53	Kotler et al. [2018]
PTEN phosphatase	Mighell et al. [2018]
amyloid β	Seuma et al. [2022]
OCT1 (SLC22A1)	Yee et al. [2023]
Tsuboyama multi-DMS	Tsuboyama et al. [2023]

Table A20: **List of indel datasets.**